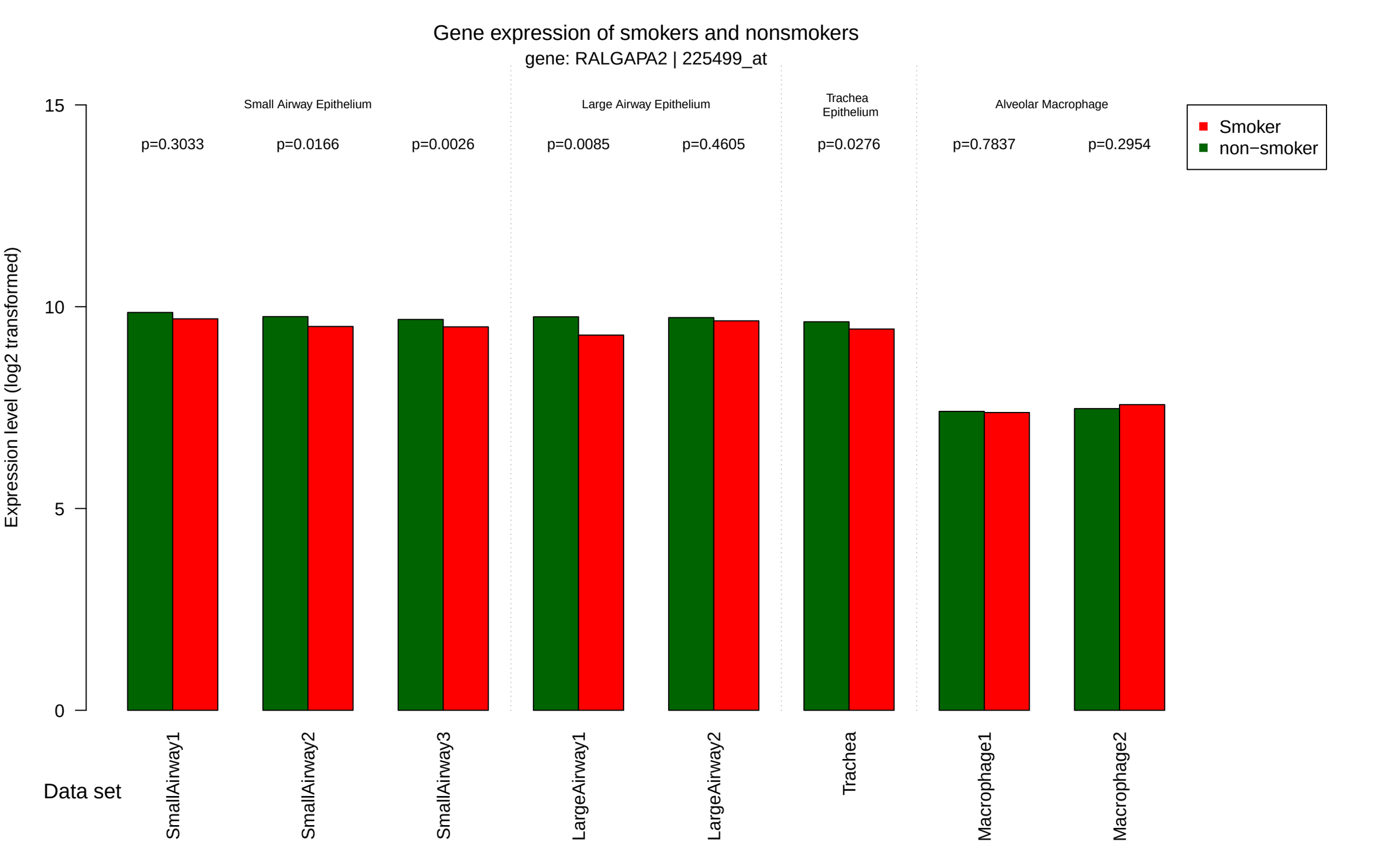
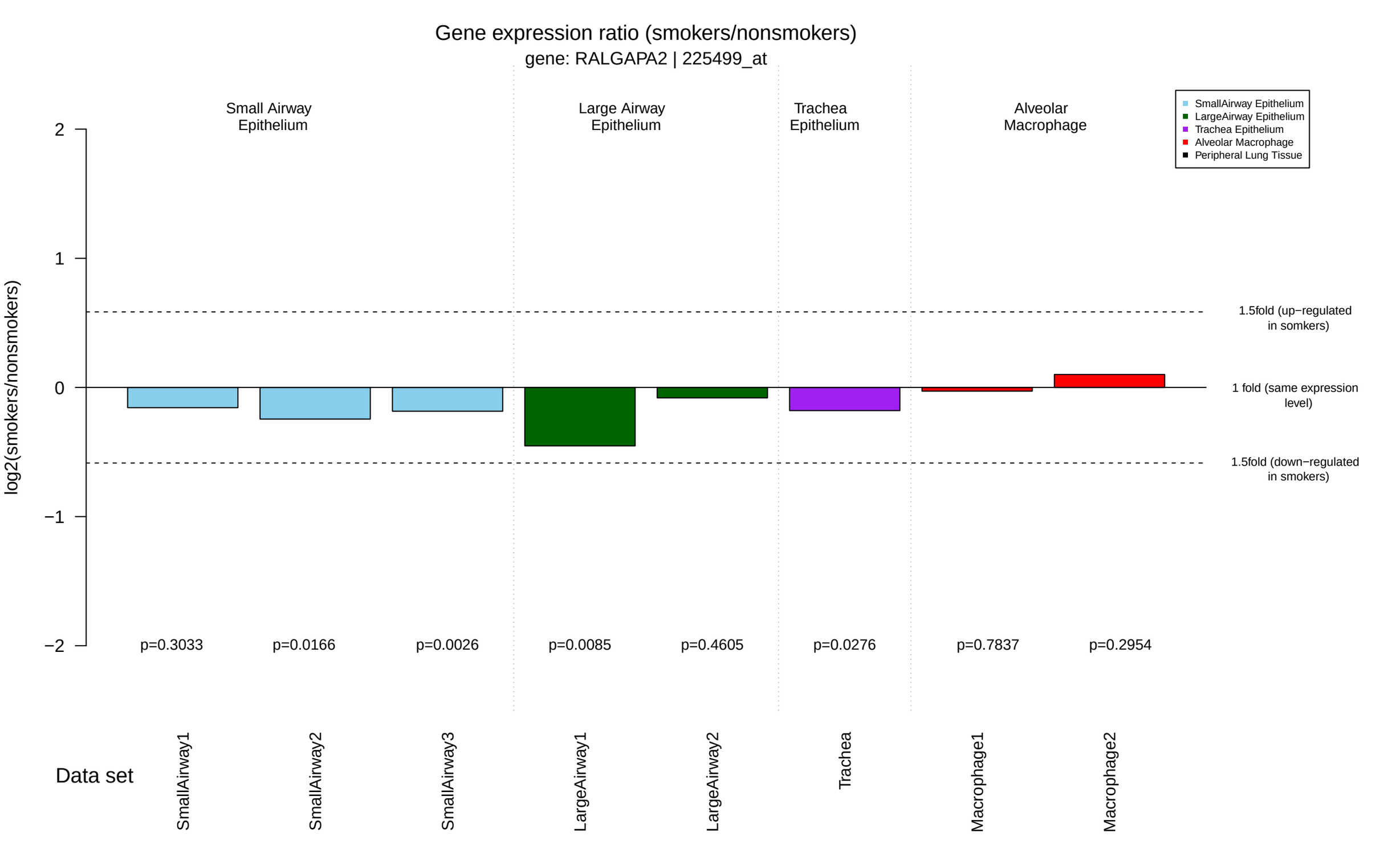
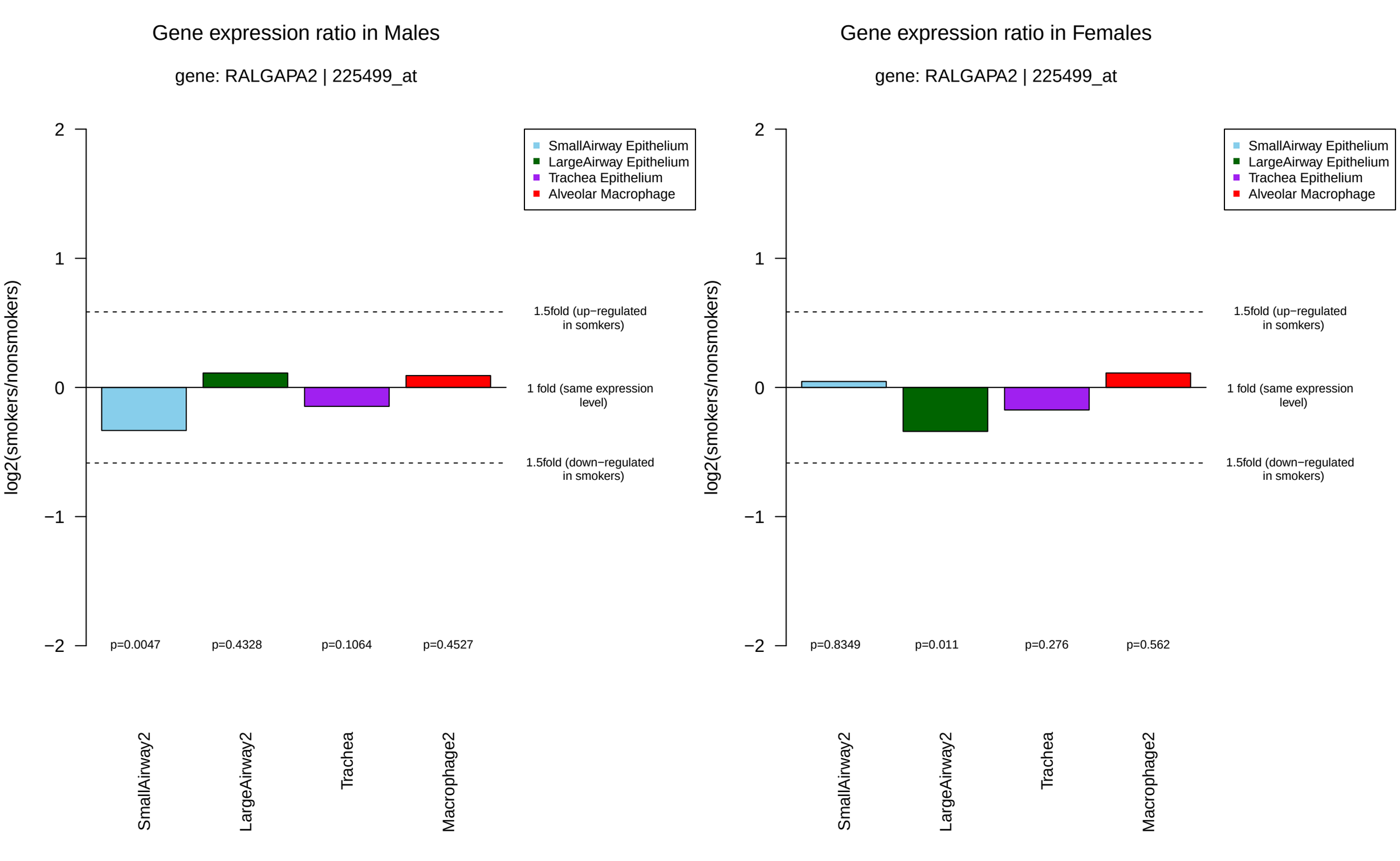
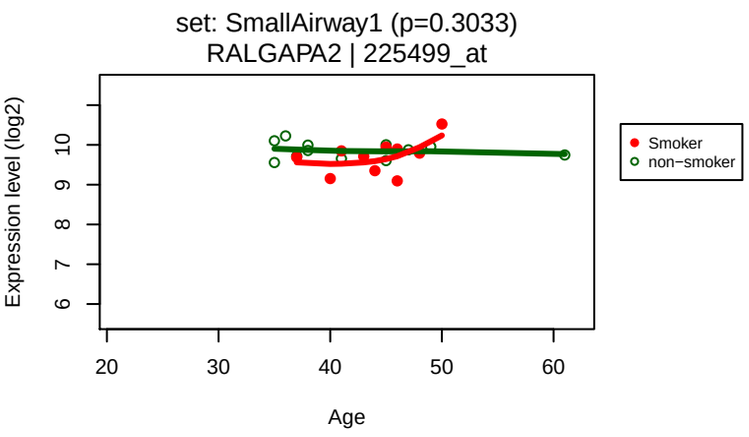
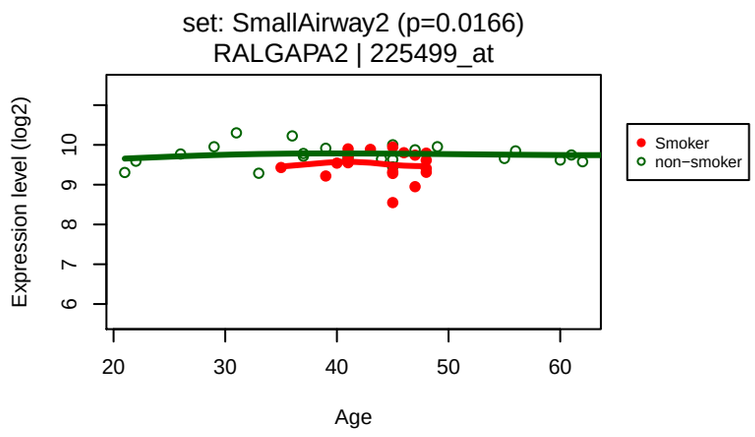
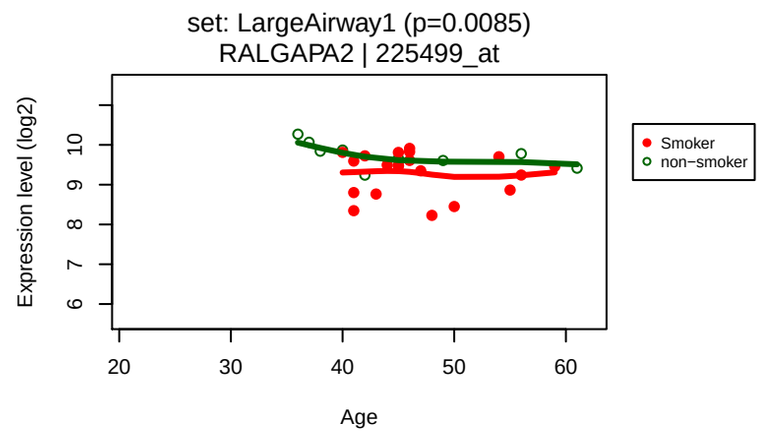

Gene expression of smokers and nonsmokers
gene: RALGAPA2 | 225499_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3033
p=0.0166
p=0.0026
p=0.0085
p=0.4605
p=0.0276
p=0.7837
p=0.2954
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

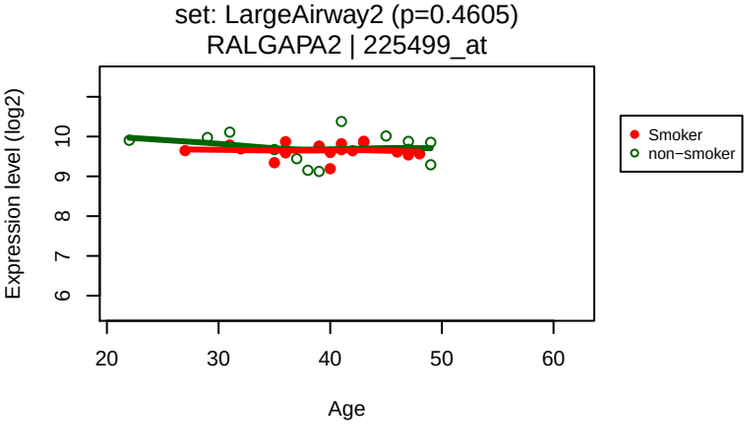

Gene expression ratio (smokers/nonsmokers)
gene: RALGAPA2 | 225499_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3033
p=0.0166
p=0.0026
p=0.0085
p=0.4605
p=0.0276
p=0.7837
p=0.2954
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

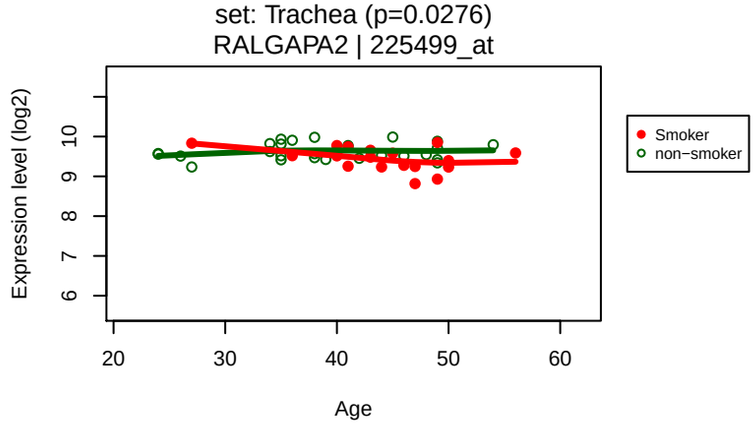

Gene expression ratio in Males
gene: RALGAPA2 | 225499_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0047
p=0.4328
p=0.1064
p=0.4527
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALGAPA2 | 225499_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8349
p=0.011
p=0.276
p=0.562
SmallAirway2
LargeAirway2
Trachea
Macrophage2

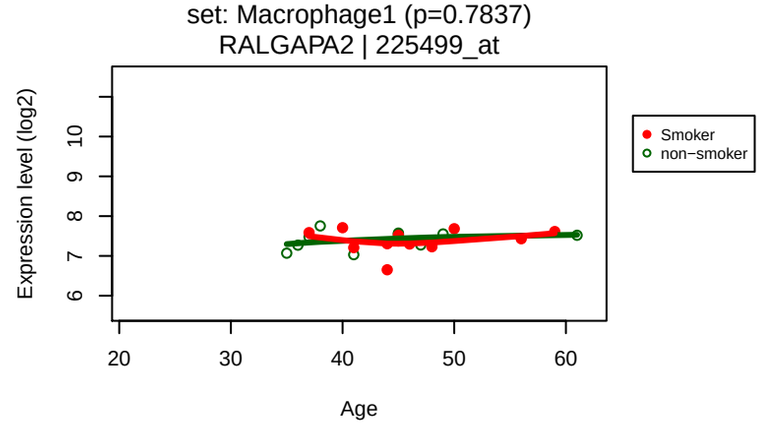

set: SmallAirway1 (p=0.3033)
RALGAPA2 | 225499_at
Expression level (log2)
Age
Smoker
non−smoker

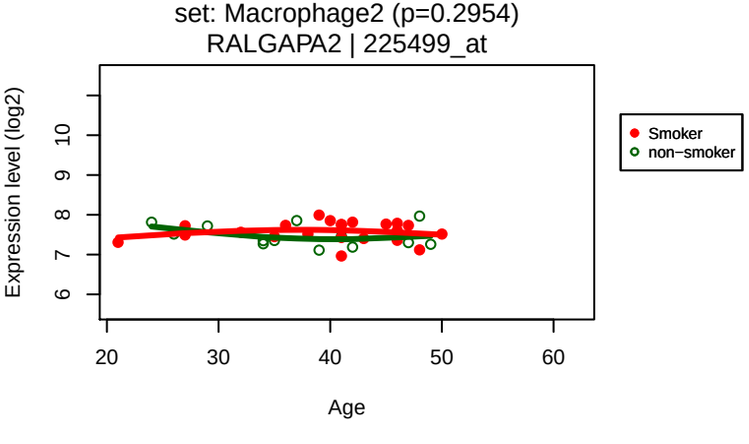

set: SmallAirway2 (p=0.0166)
RALGAPA2 | 225499_at
Expression level (log2)
Age
Smoker
non−smoker

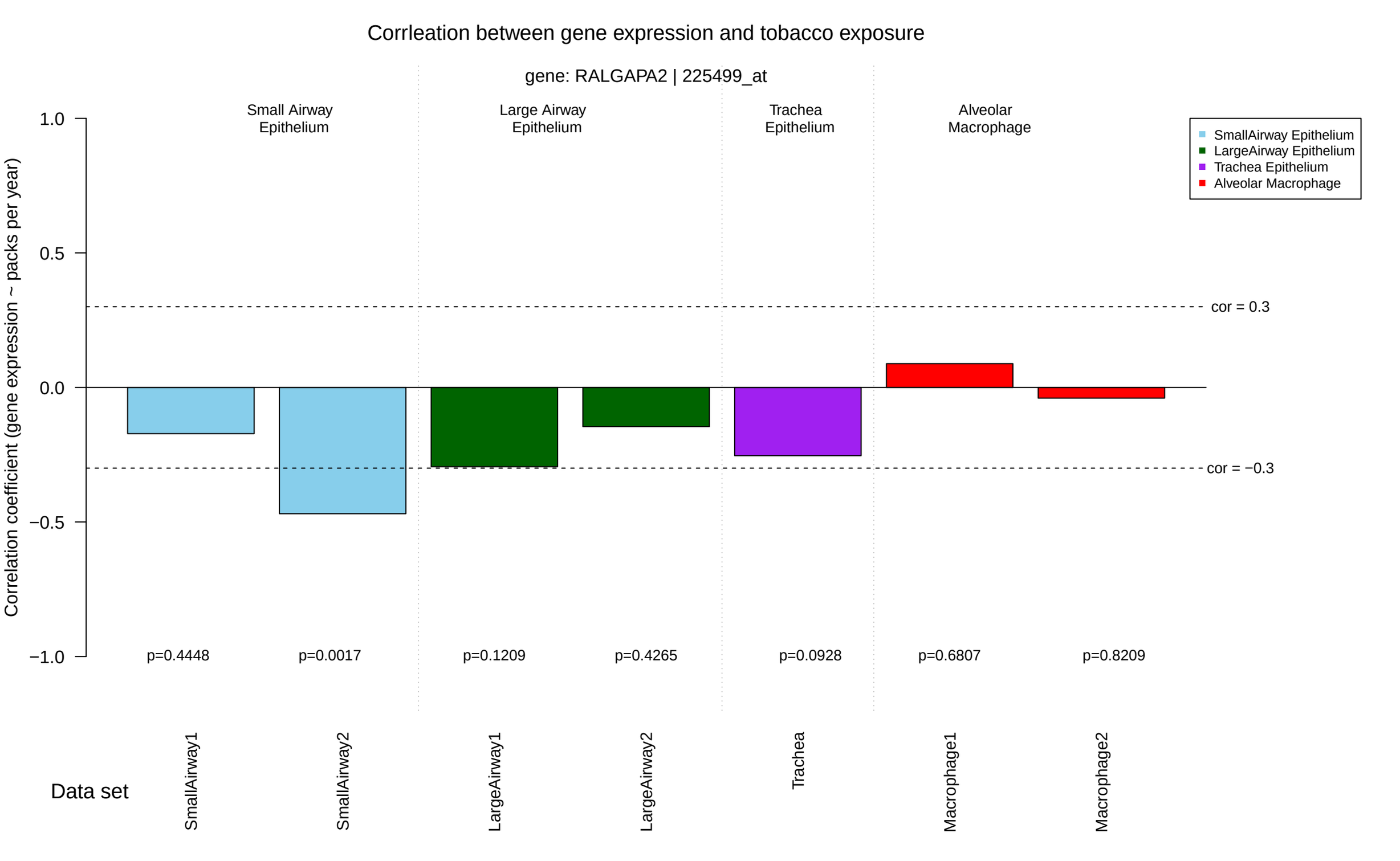

set: LargeAirway1 (p=0.0085)
RALGAPA2 | 225499_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4605)
RALGAPA2 | 225499_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0276)
RALGAPA2 | 225499_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7837)
RALGAPA2 | 225499_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2954)
RALGAPA2 | 225499_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALGAPA2 | 225499_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4448
p=0.0017
p=0.1209
p=0.4265
p=0.0928
p=0.6807
p=0.8209
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2